National Aviation Academy of the Philippines
Institute of Computer Studies

A Web-Based Service Management System with Data Analytics using Machine Learning Algorithms for Shoelotskey Villamor-Pasay

Researchers:

Angeles, Charmaine Rose G.

Gravino, Kylane C.

Sedanto, Melody D.

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Why Shoelotskey was chosen as client?

VILLAMOR - PASAY
SHOELOTSKEY
SHOE CLEANING SERVICES
Make it easy with Shoelotskey!

Shoelotskey Villamor-Pasay was selected as the client because it is a growing, established shoe care business with real operational needs that can be addressed through a digital solution.

Established Business
A recognized and operating shoe care service center in Villamor-Pasay with a growing customer base.

Clear Operational Needs
Faces actual challenges in managing job orders, inventory, and service release, making it a suitable and relevant setting for the study.

Supportive Management
The owner and staff were cooperative and open to collaboration throughout the research and development process.

Relevance to the Aviation Community
Located in an area with a strong aviation community, where proper footwear care is important for professionals who need well-maintained shoes for daily use.

INTRODUCTION

Introduction: From Manual Operations to a Data-Driven Workflow

3–10 min
Job-order retrieval
Pre-assessment baseline

2–3 days
Status-update delay
Manual updating

1–2 hrs
Report preparation
Per manual report

280–300
Monthly volume
Service orders

THE OPERATIONAL NEED

Paper-based and fragmented records made retrieval, monitoring, report consolidation, and consistent completion-date estimation more difficult.

RESEARCH GAP

Limited research integrates centralized service management, inventory monitoring, real-time analytics, and predictive modeling in one system tailored to a micro shoe-service business.

Study response → one integrated Shoelotskey-specific web platform

OBJECTIVES

Objectives of the Study

01

Manual vs Digital

Accessibility • retrieval time • accuracy

02

Random Forest

MAE • RMSE • R^2 • prediction tolerance

03

Analytics Dashboard

Reporting efficiency • information access • decisions

04

ISO/IEC 25010

Nine software-quality characteristics

05

Technology Acceptance

PU • PEOU • BI • Actual System Use

06

Future Enhancement

Evidence-based improvements and additional features

Presentation flow: Result → Discussion → Finding → Conclusion → Recommendation

RESULTS & DISCUSSION

Objective 1 — Manual vs. Digital Record Management

5.20 min
Manual mean retrieval

0.41 min
Digital mean retrieval

92.12%
Mean reduction

0 duplicate
0 incomplete
Digital pilot
31 job orders

FINDING **Centralized digital retrieval was substantially faster. The digital pilot also showed complete records under the evaluated conditions.**

DISCUSSION **Historical manual audit: 50 of 119 records (42.0%)** lacked claim/completion dates. **Digital pilot: 3 post-entry corrections (9.68%).** Accuracy findings remain descriptive because the manual and digital evidence came from different periods.

CONCLUSION → Improved accessibility, completeness, validation, and retrieval efficiency.

RESULTS & DISCUSSION

Objective 2 — Random Forest Predictive Performance

DATASET A • HISTORICAL EVIDENCE

119 validated orders from 714 paper receipts

Target: Order-to-Claim Turnaround
Issue: customer pickup behavior is included in Claim Date.

8.35 d	10.19 d	−0.0125
MAE	RMSE	R^2

Finding: historical Order-to-Claim did not directly represent actual service completion.

DATASET B • FINAL OPERATIONAL MODEL

10,000 simulated operational orders

7,000 train • 1,500 validation • 1,500 final test
Target: Received → Ready-for-Release Duration

0.0815 d	0.4042 d	0.9976
MAE	RMSE	R^2

97.20% within ±1 day • 100% within ±5 days

Conclusion → Strong predictive performance under Dataset B's simulated operational conditions.

RESULTS & DISCUSSION

Objective 3 — Analytics Dashboard Impact

43.71 min
Manual report preparation
Overall mean across 7 reports

0.0838 sec
Dashboard generation
Application/API response

7 / 7 PASS
Content verification
Independent calculations matched

3.83 / 4
Dashboard impact
Highly Acceptable • n=6

REPORTS VERIFIED

- Daily
- Weekly
- Monthly
- Quarterly
- Annual
- Expenses
- ROI

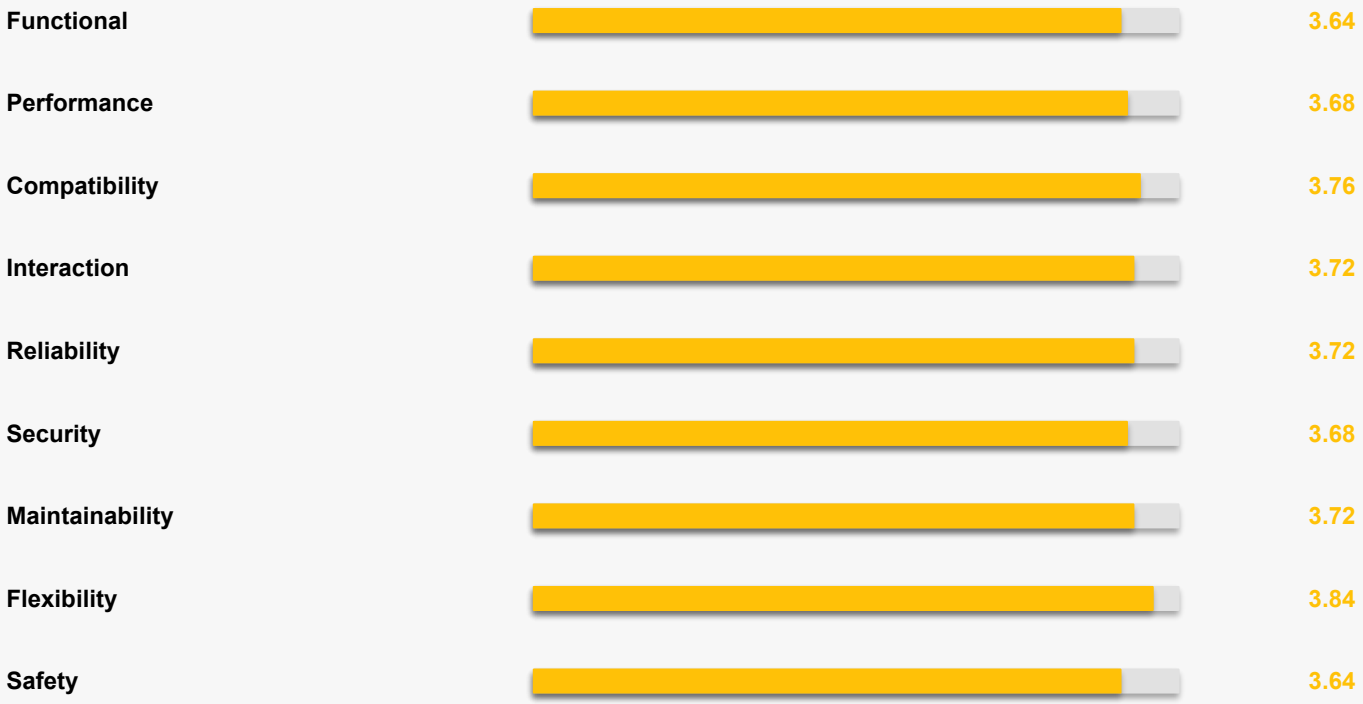
99.9968% observed reduction

Applies specifically to the evaluated report-generation stage; manual timings measured human preparation, while digital timings measured application/API processing after records were digitized.

Conclusion → Faster report generation with verified content and accessible operational information.

RESULTS & DISCUSSION

Objective 4 — Software Quality: ISO/IEC 25010



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Quality characteristics
Highly Acceptable

Highest

Flexibility
3.84

IT experts: n=5
Supported by applicable technical tests / artifact inspection.

RESULTS & DISCUSSION

Objective 5 — User Acceptance: Technology Acceptance Model

4.00 / 4	3.89 / 4	3.61 / 4	4.00 / 4
Perceived Usefulness	**Ease of Use**	**Behavioral Intention**	**Actual System Use**
Immediate usefulness	Usability	Continued-use intention	Pilot-period use

FINDING **All four TAM constructs were interpreted as Highly Acceptable among six intended users.**

DISCUSSION Perceived Usefulness and Actual System Use were highest (4.00). Behavioral Intention was lowest (3.61) but remained Highly Acceptable; the two-week pilot does not establish sustained long-term adoption.

Conclusion → Positive usefulness, usability, intention, and reported system use during the evaluation period.

SUMMARY

Summary of Findings

01

RECORDS

92.12%

Lower mean retrieval time

02

ML

0.9976 R^2

Dataset B final test

03

ANALYTICS

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reports content-verified

04

QUALITY

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ISO characteristics Highly Acceptable

05

ACCEPTANCE

4 / 4

TAM constructs Highly Acceptable

Overall: the developed system addressed the targeted operational requirements under the evaluated conditions.

CONCLUSION

Conclusion

The study achieved its primary objective of designing, developing, implementing, and evaluating the centralized Shoelotskey system.

CENTRALIZED RECORDS
Faster retrieval and structured digital information

REAL-TIME ANALYTICS
Automated reporting and operational monitoring

ML PREDICTION
Ready-for-Release Duration → ML Predicted Release Date

QUALITY & ACCEPTANCE
Highly Acceptable expert and intended-user evaluations

Evidence boundary: Dataset B demonstrates strong simulated-condition ML performance; continued actual Ready-for-Release data collection is recommended for real-world validation.

Recommendations

1	**REAL OPERATIONAL ML DATA**	Continuously capture actual For Release / Ready-for-Release timestamps; validate and retrain the Random Forest when sufficient data are available.
2	**CONFIGURABILITY**	Expand flexible dropdown options for brands, couriers, models, and other changing business entries.
3	**AUDITABILITY**	Strengthen detailed inventory movement audit logs and maintain backups, permissions, and security controls.
4	**COMMUNICATION**	Consider automated SMS/email notifications for order-status changes.
5	**ENGINEERING & LONGER USE**	Improve modularity, migrations, documentation, automated testing, and conduct longer-term evaluation with more operational data.

Q & A

Thank You!